SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION   COURSE CODE: ITA0515   Slot B

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)   Assessment Tool Weightage: 40%

QUESTIONS: 5   Total Marks: 50   Duration: 60 Minutes   Annexure A

CO4 – AT2: INDUSTRY PROBLEM SOLVING TASK (ANSWERS)

Code of Conduct:

I A SHIVA SHANKAR REDDY (192425174) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

A SHIVA SHANKAR REDDY

Signature of the student:

1. Traffic Surveillance Model Selection

Given Data

- Constraints: Accuracy ≥ 90%, Latency ≤ 80 ms, must work under varying lighting
- Method A: 88% accuracy, 60 ms latency
- Method B: 91% accuracy, 85 ms latency
- Method C: 92% accuracy, 75 ms latency

Analysis

A real-time traffic monitoring system must simultaneously satisfy an accuracy floor and a latency ceiling, since a slow model cannot keep pace with live video and an inaccurate model produces unreliable violation or incident detection. Varying lighting (daylight, night, glare, shadows) further tests the robustness of whichever model is chosen.

Method A responds fastest (60 ms), well inside the 80 ms budget, but its accuracy of 88% is below the required 90%, so it would miss more vehicles or violations than acceptable.

Method B clears the accuracy bar at 91%, but its latency of 85 ms exceeds the 80 ms limit, meaning frames would be processed too slowly for a live feed and could cause a growing backlog.

Method C is the only option that satisfies both numerical thresholds at once: 92% accuracy (≥ 90%) and 75 ms latency (≤ 80 ms).

Comparison

Factor	Method A	Method B	Method C
Accuracy	88% – Fails	91% – Passes	92% – Passes
Latency	60 ms – Passes	85 ms – Fails	75 ms – Passes
Meets both constraints	No	No	**Yes**
Real-time suitability	High	Low	High
Overall verdict	Rejected (low accuracy)	Rejected (high latency)	Selected

Evaluation

Because traffic surveillance decisions can trigger enforcement or safety actions, both constraints should be treated as hard requirements rather than soft targets. Method A's speed advantage does not compensate for a 2-point accuracy shortfall, since missed detections directly reduce system reliability. Method B's accuracy margin is the smallest of the compliant options and it still breaches the latency ceiling, which risks frame queuing on a live stream.

Method C is the only method that clears both thresholds, and does so with reasonable margin (2 points of accuracy headroom and 5 ms of latency headroom), which also gives it some tolerance for the additional processing load introduced by lighting-compensation techniques.

Conclusion

Method C should be selected for deployment, as it is the only method that simultaneously satisfies the accuracy and latency constraints. Its robustness under varying lighting should still be validated using techniques such as adaptive histogram equalization, HDR imaging, or training on illumination-augmented datasets before final deployment.

2. Drone Vision System Constraint

Given Data

- Requirements: lightweight computation, real-time detection (< 60 ms), accuracy ≥ 85%
- Model A: 83% accuracy, 40 ms latency
- Model B: 87% accuracy, 65 ms latency
- Model C: 86% accuracy, 55 ms latency

Analysis

Drones operate under tight resource budgets – limited battery, onboard compute, and payload – and depend on fast, reliable detections to support flight-control decisions such as obstacle avoidance or target tracking.

Model A is the fastest at 40 ms, comfortably inside the 60 ms limit, but its accuracy of 83% falls short of the 85% requirement, which risks missed obstacles or targets during flight.

Model B has the highest accuracy at 87%, but its 65 ms latency exceeds the 60 ms real-time limit. A delayed detection at flight speed can translate directly into a collision risk.

Model C meets both numerical requirements: 86% accuracy (≥ 85%) and 55 ms latency (< 60 ms).

Comparison

Factor	Model A	Model B	Model C
Accuracy	83% – Fails	87% – Passes	86% – Passes
Latency	40 ms – Passes	65 ms – Fails	55 ms – Passes
Meets both constraints	No	No	**Yes**
Implied compute load	Lowest	Highest	Moderate
Overall verdict	Rejected (low accuracy)	Rejected (too slow)	Selected

Evaluation

For drones, latency carries the same safety weight as accuracy, since a delayed reaction at flight speed can be as dangerous as a missed detection. Model A's speed does not offset its insufficient accuracy for critical detection tasks. Model B's superior accuracy is undermined by breaching the real-time limit, making it unsuitable for onboard, in-flight use.

Model C is the only model satisfying both constraints, and its shorter inference time relative to Model B also suggests a comparatively leaner network, which aligns with the lightweight-computation requirement expected on embedded drone hardware.

Conclusion

Model C is the preferred choice for drone deployment, as it is the only option meeting both the real-time and accuracy constraints while implicitly fitting the lightweight-computation requirement. Model B could still be useful for offline or ground-station analysis where the 60 ms limit does not apply.

3. Security System Recognition

Given Data

- Requirements: high accuracy (> 92%), moderate computation
- Traditional: 85% accuracy, low computation
- Deep Learning: 94% accuracy, high computation
- Hybrid: 91% accuracy, moderate computation

Analysis

This case illustrates the classic accuracy-versus-resource trade-off common in recognition systems. No single option satisfies both stated constraints simultaneously, which makes this a judgment-based, not a purely numerical, decision.

Traditional methods are computationally cheapest but their accuracy of 85% falls well short of the 92% security threshold, which is risky for identity verification or access-control decisions.

Deep Learning comfortably exceeds the accuracy target at 94%, but its computational demand is high, which can conflict with the moderate-computation constraint unless additional hardware (GPU/cloud inference) is available.

Hybrid keeps computation at the required moderate level, but its accuracy of 91% sits just under the > 92% requirement.

Comparison

Factor	Traditional	Deep Learning	Hybrid
Accuracy	85% – Fails	94% – Passes	91% – Fails (close)
Computation	Low	High – exceeds 'moderate'	Moderate – Passes
Gap from target	-7 points	+2 points	-1 point
Best suited when	Low-security, low-cost setups	Accuracy is non-negotiable	Hardware/cost is constrained

Evaluation

Traditional methods can be ruled out immediately – an 7-point accuracy shortfall is too large for a security application where false negatives have real consequences.

Between Deep Learning and Hybrid, the choice depends on which constraint the deployment can least afford to violate. If accuracy is non-negotiable (e.g., high-security access control, surveillance of critical infrastructure), Deep Learning is preferable despite the higher hardware cost, because a misidentification is costlier than the extra computation. If the deployment environment has strict hardware, power, or cost limits (e.g., embedded security cameras), Hybrid is the more practical option, since it is only 1 point below the accuracy target while already meeting the computation constraint.

Conclusion

There is no option that satisfies both constraints exactly. Deep Learning is recommended when accuracy is the overriding priority and adequate computing resources are available; Hybrid is the pragmatic choice when computation must remain moderate. Traditional methods are not suitable for high-security requirements in either case.

4. Motion Tracking in Sports Analytics

Given Data

- Requirements: accurate motion tracking, real-time performance
- Optical Flow: accurate, slow
- Motion Estimation: fast, less accurate

Analysis

Optical Flow computes dense, pixel-level motion vectors between consecutive frames, producing highly accurate motion trajectories that are valuable for detailed biomechanical or tactical analysis. This precision comes at the cost of heavy computation, which makes it slow and unsuitable for live processing.

Motion Estimation techniques (e.g., block-based matching or sparse feature tracking) run much faster and are well suited to real-time overlays or broadcast graphics, but they sacrifice tracking precision, which can introduce jitter or less reliable trajectory data.

Comparison

Factor	Optical Flow	Motion Estimation
Tracking accuracy	High	Lower
Processing speed	Slow	Fast
Real-time suitability	Low	High
Best use case	Post-match / offline analysis	Live broadcast overlays

Evaluation

The right choice depends on when the analysis happens, not on which method is objectively "better." For real-time use – live match broadcasts, on-field coaching feedback, or interactive player-tracking overlays – Motion Estimation is preferable because responsiveness matters more than pixel-level accuracy, and minor inaccuracies are acceptable for visualization purposes.

For offline or analytical use – post-match performance review, biomechanics, or injury-risk assessment – Optical Flow is preferable because precision is critical and there is no real-time processing constraint.

Conclusion

Neither method is universally superior; Motion Estimation suits real-time sports broadcasting while Optical Flow suits detailed offline analytics. A practical pipeline could combine both – fast Motion Estimation for live overlays and Optical Flow refinement afterward for accurate post-game analysis.

5. Face Recognition System Design

Given Data

- Constraints: Accuracy ≥ 90%, works in low light, moderate computation
- Traditional: 82% accuracy
- Deep Learning: 93% accuracy
- Hybrid: 89% accuracy

Analysis

Traditional handcrafted-feature methods (e.g., Eigenfaces, LBP) reach only 82% accuracy, well below the 90% requirement, and such methods are also known to degrade sharply in low-light conditions because they depend on clearly defined edges and textures.

Deep Learning reaches 93% accuracy, comfortably clearing the threshold, and CNN-based models trained with low-light or illumination-augmented data generalize noticeably better to poor lighting than handcrafted approaches. The trade-off is that deep networks typically require more computation than a "moderate" budget allows, unless optimized.

Hybrid reaches 89% accuracy – close to, but just under, the 90% requirement – while offering a more balanced, moderate computation profile.

Comparison

Factor	Traditional	Deep Learning	Hybrid
Accuracy	82% – Fails	93% – Passes	89% – Fails (close)
Low-light robustness	Poor	Good (with training)	Moderate
Computation	Low	High (needs optimization)	Moderate – Passes
Gap from accuracy target	-8 points	+3 points	-1 point

Evaluation

Traditional methods can be eliminated immediately – they miss the accuracy target by 8 points and lack low-light robustness, failing two of the three constraints outright.

Deep Learning satisfies the two hardest constraints – accuracy and low-light performance – but must be engineered (lightweight backbones such as MobileFaceNet, pruning, or quantization) to bring its computational footprint down to a moderate level suitable for the target hardware.

Hybrid narrowly misses the accuracy bar (89% vs. the required 90%) but already matches the moderate-computation requirement, making it a reasonable fallback rather than the primary choice.

Conclusion

Since accuracy and low-light performance are safety- and security-critical for face recognition (e.g., authentication, access control), Deep Learning is the recommended choice once optimized for computational efficiency, as it is the only option that clears both the accuracy and low-light requirements. Hybrid remains a viable fallback for resource-constrained deployments where the small accuracy gap is acceptable.

Overall Conclusion

Across the five cases, model or method selection under real-world constraints rarely comes down to picking the single most accurate option. Each scenario requires checking every stated constraint – accuracy, latency, computation, and environmental robustness – rather than optimizing one metric in isolation.

- For traffic surveillance and drone vision, only one option (Method C and Model C respectively) satisfied every hard numerical constraint simultaneously, making the decision largely deterministic.
- For the security and face-recognition systems, no option satisfied every constraint exactly, so the decision depended on which requirement the deployment could least afford to compromise, favouring Deep Learning when accuracy is critical and hardware can be optimized.
- For sports-analytics motion tracking, the trade-off was purely contextual (real-time versus offline use), showing that some decisions depend on the deployment phase rather than fixed thresholds.

Overall, effective computer vision system design requires evaluating every candidate against the complete set of application constraints – accuracy, speed, computation, and environmental robustness – and making a justified, context-driven trade-off rather than defaulting to the highest-accuracy option.

IMAGE REPRESENTATION